The Two-Face Problem:

A Practical Guide to Translating Binary-Partition Formulas Across Disciplines

Elias De Jesús
Independent Researcher
ORCID: 0009-0007-0190-9143 dejesuselias10@gmail.com

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Abstract

Many disciplines use formulas that look structurally related but behave differently, because they are written in different variables, on different domains, or under different boundary conditions. This guide gives a practical way to diagnose such cases when the quantities involved form a two-channel budget $a+b=1$. Every such budget lies on the same semicircle (the Thales arc), $(a-b)^2+4ab=(a+b)^2$, and the guide supplies an exact dictionary between the coordinates commonly used on it: the bounded asymmetry and Fisher angle, the unbounded rapidity (half the log-odds), and the radial response coordinate; the bounded angle and the rapidity are linked by the Gudermannian.

Revision 3 reorganizes the guide around one principle: *the arc is universal, but how it is read is supplied by the mechanism.* An *instantiation* fixes, from the domain and before any comparison, what a and b are; whether the budget is actually closed or leaks; whether the coupling $\sqrt{ab}$ is realized or only potential; which coordinate is native, judged by the domain's own composition law; and whether orientation matters. Translations between coordinates are exact but, applied to a known formula, they are normal forms rather than discoveries; a reading earns empirical content only by passing an over-determination test (a family fit, an independently determined law, cross-domain recurrence under a fixed normalization, or compression). A worked analysis of the SXS binary-black-hole catalog shows the procedure at work: the coordinate chosen from the physics (the symmetric mass ratio $\eta=h^2$) gives the most compact description of radiated energy and final spin out of sample; the radiated-energy leakage is consistent with, though not a sharp test of, the test-particle limit (ISCO binding energy times the coupling); and the recoil is compactly written as coupling times asymmetry, reproducing the known maximum near 175 km/s.

The empirical "equipartition differential" reported in Revision 2 is withdrawn: a null simulation reproduces it from unequal bin widths alone, and it does not replicate on the current SXS catalog. The heuristic sections on a "dimensional selection" from a powered metric family and on "Pythagorean power distributions" are removed. An error in one worked translation (the generating function $G(t)$) is corrected.

Keywords: binary partition; Thales arc; Gudermannian; Fisher metric; rapidity; exponential family; leakage; instantiation; normal form; falsifiability; SXS catalog.

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1 The problem, and what this revision changes

Across the sciences one meets formulas that look related but behave differently. A normalized occupation is bounded, a Boltzmann factor is not; a Lorentz factor is hyperbolic, an inverse-square law is radial. This guide addresses a limited version of that observation. When the quantities form a conserved two-channel budget $a + b = 1$, the budget leaves one degree of freedom, and several familiar formulas turn out to be the same state written in different coordinates. Translating between those coordinates can show that an apparent mismatch is only a change of variables, or that it is a real difference that survives translation. Both outcomes are useful.

What Revision 3 changes. Revision 2 presented three fixed "descriptions" of the partition state (Fisher, Poincaré, Compound) and chose among them by which one made a formula look simplest. That criterion is subjective, and it cannot distinguish a coordinate that is merely available from one the domain actually uses. Revision 3 replaces it with three ideas developed since:

1. *Instantiation* (Section 4): the arc is a universal constraint locus, and the domain supplies how it is read — embedding, closure, coupling type, native coordinate, orientation, dynamics — fixed before comparison.

2. *Normal forms versus tests* (Section 6): an exact translation of a known formula is a normal form; a claim acquires content only by passing an over-determination test [1].
3. *Leakage* (Section 5): when the budget is not closed, the state lives in a half-disk rather than on the arc [2, 3].

The changes in the empirical and heuristic material are listed in Section 10.

Epistemic tiers. **Tier 1**: exact identities and translations. **Tier 2**: readings that have passed an over-determination test in a stated setting, or empirical results in a stated dataset. **Tier 3**: interpretive or speculative claims. Following [4], a Tier-1 translation is never by itself evidence for shared physics.

Quick start. (1) Identify the two channels and check whether their total is conserved. (2) Instantiate the arc from the mechanism (Section 4) *before* looking at data. (3) Translate with the dictionary (Section 3). (4) Classify the result: exact translation, partial correspondence, mixed object, or mismatch (Section 7). (5) If you want more than a re-expression, run an over-determination test (Section 6) and report it, including failures.

2 The arc

2.1 The binary partition and its coordinates

A two-channel budget with shares

$$a + b = 1, \qquad a, b \in (0, 1) \tag{1}$$

leaves one degree of freedom. We use the *signed* asymmetry

$$\chi = a - b \in (-1, 1), \qquad a = \tfrac{1+\chi}{2}, \quad b = \tfrac{1-\chi}{2}. \tag{2}$$

Revision 2 used $|2a-1|$. The signed form is preferred because many mechanisms distinguish the two channels (Section 4); symmetric quantities below are even in χ and cannot do so. Define

$$h = \sqrt{ab}, \qquad L = \frac{a-b}{2}, \qquad R = \frac{1}{1-\chi^2} = \frac{1}{4h^2}, \qquad \xi = \operatorname{artanh}\chi, \qquad \theta = \arcsin\chi, \tag{3}$$

the altitude (geometric mean), half-asymmetry, response, rapidity and Fisher angle.

Proposition 1 (The Thales arc; Tier 1)**.** *For any nonnegative a, b,*

$$(a-b)^2 + 4ab = (a+b)^2. \tag{4}$$

Hence on a closed budget $L^2 + h^2 = \frac{1}{4}$ and $\chi^2 + (2h)^2 = 1$: every binary partition lies on the same semicircle of diameter $a + b$.

The proof is one line of algebra. The consequence matters for interpretation: lying on the arc is automatic, so it is never evidence that two domains share structure. The name refers to Thales' theorem: h is the altitude of the right triangle inscribed in the semicircle of diameter $a + b$.

2.2 The mean hierarchy

The arithmetic, geometric and harmonic means of the shares are $\frac{1}{2}$, h and $2h^2$.

Proposition 2 (Adjacent-gap ratio; Tier 1)**.**

$$\frac{\mathrm{AM} - \mathrm{GM}}{\mathrm{GM} - \mathrm{HM}} = \frac{1/2 - h}{h - 2h^2} = \frac{1}{2h} = \frac{1}{\sqrt{1-\chi^2}}.$$

2.3 Two dual coordinates: shares and forces

For a binary random variable with success probability a, the share a (equivalently χ) is the *mean* parameter of the Bernoulli family, and the log-odds $\log(a/b) = 2\xi$ is its *natural* parameter. The two are related by

$$\chi = \tanh \xi, \qquad \frac{d\xi}{d\chi} = \frac{1}{1-\chi^2} = R, \tag{5}$$

the Legendre correspondence of the exponential family [5]. This is the standard reason a bounded coordinate and an unbounded one appear together. The bounded face records *how much* of the budget each channel holds. The unbounded face records the conjugate *force*: a chemical-potential difference, a log-likelihood ratio, an impedance ratio, or a rapidity. The response R is the exchange rate between them: the force needed per unit change of share. It grows without bound near the boundary, which is the precise sense in which pushing an already lopsided split further is costly.

3 Metrics and the translation dictionary

3.1 Three metrics on one line

Three metrics on $\chi \in (-1, 1)$ recur in applications. Each has a flat (arc-length) coordinate:

$$ds_F^2 = \frac{da^2}{a} + \frac{db^2}{b} = \frac{d\chi^2}{1-\chi^2} = d\theta^2, \qquad \theta = \arcsin \chi \in (-\tfrac{\pi}{2}, \tfrac{\pi}{2}), \tag{6}$$

$$ds_P^2 = \frac{d\chi^2}{(1-\chi^2)^2} = d\xi^2, \qquad \xi = \operatorname{artanh} \chi \in \mathbb{R}, \tag{7}$$

$$ds_C^2 = \frac{\chi^2\, d\chi^2}{(1-\chi^2)^4} = dy^2, \qquad y = \tfrac{R}{2} \in [\tfrac{1}{2}, \infty). \tag{8}$$

The first is the Fisher metric of the Bernoulli family [6, 7]. The second is the one-dimensional hyperbolic (rapidity) metric. The third is the radial metric of Revision 2; it is degenerate at $\chi = 0$, so y is a chart on each half $\chi > 0$ and $\chi < 0$ separately. They are related exactly by

$$ds_C^2 = J^2 \gamma^2\, ds_P^2, \qquad J = \sinh \xi, \quad \gamma = \cosh \xi. \tag{9}$$

Remark 1 (What distinguishes them; Tier 1). Every one-dimensional metric is flat, so the three differ not in curvature but in two things: whether the boundary $|\chi| \to 1$ lies at finite distance (Fisher) or infinite distance (rapidity, radial), and which coordinate is affine. Which metric is physically relevant is supplied by the domain (Section 4). A kinetic term written with the Fisher denominator and expressed in ξ carries a residual $\operatorname{sech}^2 \xi$ weight; that weight reflects the metric–coordinate pairing, not domain physics [1].

3.2 The Gudermannian bridge and the dictionary

The bounded angle and the rapidity are linked by the Gudermannian:

$$\begin{gathered} \theta = \operatorname{gd} \xi = \arctan(\sinh \xi), \\ \sin\theta = \tanh\xi = \chi, \quad \cos\theta = \operatorname{sech}\xi = 2h, \quad \tan\theta = \sinh\xi, \quad \sec\theta = \cosh\xi. \end{gathered} \tag{10}$$

A useful companion is the square-root (Bhattacharyya) angle $\varphi = \arcsin\sqrt{a}$, which embeds the partition on a circle as $(\sqrt{a}, \sqrt{b}) = (\sin\varphi, \cos\varphi)$. It satisfies $\theta = 2\varphi - \pi/2$, and the Fisher distance between two partitions is twice the difference of their φ values.

Table 1: The translation dictionary on a closed budget ($\chi \geq 0$ branch shown for the radial column; for $\chi < 0$ odd quantities change sign). All entries are exact (Tier 1) and are verified in `verify_rev3.py`.

Quantity	Fisher angle θ	Rapidity ξ	Radial $y = R/2$
Asymmetry χ	$\sin\theta$	$\tanh\xi$	$\sqrt{1 - 1/(2y)}$
Altitude h	$\frac{1}{2}\cos\theta$	$\frac{1}{2}\operatorname{sech}\xi$	$1/\sqrt{8y}$
$J = \chi/\sqrt{1-\chi^2}$	$\tan\theta$	$\sinh\xi$	$\sqrt{2y-1}$
$\gamma = 1/(2h)$	$\sec\theta$	$\cosh\xi$	$\sqrt{2y}$
Response R	$\sec^2\theta$	$\cosh^2\xi$	$2y$
Log-odds $\log(a/b)$	$2\,\operatorname{artanh}(\sin\theta)$	2ξ	$2\,\operatorname{arccosh}\sqrt{2y}$

4 Instantiating the arc

The arc is the same for every two-channel budget. What changes from one domain to another is everything that gives the arc meaning, and none of it comes from the geometry. We call a specification of those components an *instantiation*.

Definition 1 (Instantiation)**.** An instantiation of the arc for a mechanism M fixes, from M alone:

(i) **Embedding**: what the channels a and b are physically;

(ii) **Closure**: whether $a + b$ is conserved, or whether a third channel takes part of the total (Section 5);

(iii) **Coupling type**: whether $2h = 2\sqrt{ab}$ is the realized coupling between the channels, or only the maximum possible coupling (Section 5);

(iv) **Native coordinate**: the coordinate in which the domain's own composition law is additive, or in which its dynamics close;

(v) **Orientation**: whether M distinguishes a from b, in which case the signed χ (or ξ) is required and the symmetric quantities h, R, y are insufficient;

(vi) **Dynamics and landmarks**: what moves the state, and which points M itself singles out.

The pre-commitment rule. An instantiation must be fixed by the mechanism *before* anything is compared with data or with a target value. Chosen afterwards, the components carry enough freedom to fit almost anything; this is the threshold-manufacturing risk of [4]. With the rule in place, every claim has the form "under the instantiation supplied by M, we expect X." If X fails, that instantiation is rejected and the arc is untouched.

4.1 The native coordinate is decided by composition

Revision 2 chose a "native description" by simplicity of form. Revision 3 uses an operational criterion: *a coordinate is native to a domain when the domain's own way of combining two states becomes addition in that coordinate.* Examples:

- **Collinear Lorentz boosts.** Einstein addition $\beta_1 \oplus \beta_2 = (\beta_1 + \beta_2)/(1 + \beta_1\beta_2)$ is addition of rapidities and multiplication of the odds $(1+\beta)/(1-\beta)$, which are the squared Bondi k-factors [8–10]. Rapidity is native.
- **Bayesian updating of a binary hypothesis.** Independent evidence multiplies likelihood ratios, so log-odds $= 2\xi$ add. Rapidity is native.

- **Resistive impedance mismatch.** For a real load Z on a line of impedance Z_0, the reflection coefficient is $\Gamma = (Z - Z_0)/(Z + Z_0) = \tanh\big(\frac{1}{2}\ln(Z/Z_0)\big)$. Changing reference impedance multiplies impedance ratios, so $\xi = \frac{1}{2}\ln(Z/Z_0)$ adds. Rapidity is native, and the standing-wave ratio is $e^{2|\xi|}$.
- **Cosmic expansion.** Scale factors multiply, so e-folds $N = \ln a$ add. In the flat matter–Λ partition with $a = \Omega_\Lambda$, the rapidity flows at the constant rate $d\xi/dN = +\frac{3}{2}$ (Section 8).
- **Estimation of a proportion.** Here there is no composition of states, and the native coordinate is fixed instead by the statistics: the Fisher (or Bhattacharyya) angle, in which the estimation noise is uniform. The estimator $\arcsin\sqrt{\hat{p}}$ has variance close to $1/(4n)$ whatever p is (variance stabilization). The Fisher information $8y$ per trial is the metric weight in the share coordinate, not a composition law.

Where no composition law is available, the domain's dynamics or its standard expansions decide. The SXS analysis of Section 9 is an example: there the native coordinate is the coupling $h^2 = \eta$, the expansion variable of post-Newtonian theory.

4.2 Constraint types

The kind of constraint a mechanism imposes tells you which coordinate to expect:

Table 2: Constraint types and the coordinates they act on (Tier 3: an organizing heuristic, to be checked for each mechanism).

Constraint	How it acts	Expected coordinate	Examples
Scarcity / budget	fixes the total $a + b$	shares a, b or χ	probabilities, mass fractions
Pressure / force	sets the position on the arc	rapidity ξ	Fermi–Dirac (log-odds of occupation $= -(E - \mu)/kT$), voltages
Conservation	total fixed while the split changes	motion along the arc	energy exchange $T \leftrightarrow V$
Dissipation / friction	drives the split to a rest point	a flow on the arc	relaxation; near equilibrium, flux $\propto$ force [11]

5 Leakage and partial coupling

5.1 Open budgets

Suppose a third channel $Q \geq 0$ takes part of a total E: $T + V + Q = E$ for two active channels $T, V \geq 0$, with leakage fraction $\lambda = Q/E$ [2, 3]. (We write λ here to avoid a clash with the half-asymmetry L of (3).) Write $a = T/E$, $b = V/E$, so $a + b = 1 - \lambda$.

Proposition 3 (Angle and radius; Tier 1)**.** *With conditional shares $A = a/(1 - \lambda)$, $B = b/(1 - \lambda)$,*

$$(A - B)^2 + 4AB = 1, \qquad (a - b)^2 + 4ab = (1 - \lambda)^2, \qquad R_{\text{act}} = \frac{1}{4ab} = \frac{R_{\text{cond}}}{(1 - \lambda)^2}.$$

So the state is a point in a half-disk. The *angle* records the internal exchange between the two active channels. The *radius* $1-\lambda$ records how much of the total remains active. Leakage that draws on both channels in proportion moves the point inward without changing its direction; leakage from one channel also changes the angle.

Two practical consequences follow. First, the closure question (ii) of Definition 1 must be asked of every application: if the budget is not closed, normalizing to $a + b = 1$ hides a second variable. Second, the global response R_{act} grows as the active fraction shrinks even when nothing changes internally, so a large observed response can be produced by leakage alone.

A familiar geometry. With amplitudes $x = \sqrt{a}$, $y = \sqrt{b}$ one has $(a - b, 2\sqrt{ab}) = (x^2 - y^2, 2xy)$ and $a + b = x^2 + y^2$: the arc is the image of a quarter-circle of amplitudes under squaring. This is the geometry of the Stokes parameters of two field components [12] and of the Bloch vector of a two-level system [13]. The leakage picture is the same bookkeeping used for population leaking out of the computational subspace of a qubit [14].

5.2 Realized versus potential coupling

In optics and two-level systems the vector can also move inward *without* leakage, when the two channels lose phase coherence while their populations stay fixed. The realized coupling is then smaller than $\sqrt{ab}$. Because h is defined as $\sqrt{ab}$, it is the *maximum* coupling the shares allow, and it cannot detect such dephasing. This is question (iii) of Definition 1: whether $2h$ describes an actual interaction or only an upper bound depends on the mechanism.

5.3 Scale as a hidden coordinate

When a family test fails because the extracted dynamics depend on the total $S = a + b$, the correct conclusion is that the state is two-dimensional (share and scale). The worked example is one-loop gauge-coupling running: the extracted potential scales as S^2, so no autonomous one-dimensional partition potential exists [1]. Leakage and gauge running are therefore two instances of the same closure question.

6 Normal forms and over-determination tests

Remark 2 (Why translation alone cannot fail)*.* Translating a known formula into another coordinate is exact and always succeeds. In the same way, extracting an effective potential from one known trajectory reconstructs that trajectory exactly, because the free function absorbs the input [1]. Such results are *normal forms*: useful for classification, but not evidence of a mechanism.

A reading of the arc acquires empirical content only when it is asked to do something it was not built to do. Four routes, from [1]:

1. **Family or held-out test**: one law must fit a whole family of cases, including cases not used to fix it, with the residual reported.
2. **Independent determination**: a principle from the domain (an action, a free energy, a conservation law, a limiting case) fixes the law, which then predicts the data.
3. **Cross-domain recurrence under a fixed normalization**, declared before the comparison.
4. **Compression**: a law with few parameters replaces a long description.

Each route can fail, and a failure is informative: it classifies the domain (for example, as not governed by an autonomous one-dimensional law) rather than refuting the arc.

7 Worked translations

The pairs below are exact translations (Tier 1). Each is a normal form in the sense of Remark 2. The "native" column states the coordinate supplied by the domain's composition law where one exists; it is not a claim of physical equivalence between the domains in a row.

Table 3: Worked translations. Formulas in Appendix A.

Pair	Exact statement	Native coordinate and reason
Fermi–Dirac / logistic	$1/(e^x+1) = (1-\tanh(x/2))/2$; logit $= 2\xi$	rapidity: log-odds add under independent evidence [15, 16]
Mercator / Gudermannian	latitude $\phi_{\text{lat}} \mapsto \text{gd}^{-1}\,\phi_{\text{lat}}$	the bridge itself [17]
Bernoulli Fisher information	$I_F = 1/(ab) = 1/h^2 = 4R = 8y$	Fisher/Bhattacharyya angle: noise is uniform there [6, 18]
Relativistic aberration	$\tan\frac{\theta'}{2} = e^{-\xi}\tan\frac{\theta}{2}$	rapidity: boosts compose additively [19]
VSWR	$\text{VSWR} = (1+\lvert\Gamma\rvert)/(1-\lvert\Gamma\rvert) = e^{2\lvert\xi\rvert}$	rapidity, for resistive loads: impedance ratios multiply [20]
Altitude integrals	$\int_0^1 h^\alpha dx = B(\frac{\alpha}{2}+1, \frac{\alpha}{2}+1)$	none needed: a definite integral is invariant under change of variables
Generating function	$G(t) = \int_0^1 \frac{dx}{1-t\,x(1-x)} = \frac{\varphi}{h}$, $a = t/4$	corrected in this revision (App. A)
Langevin function	$\coth x - 1/x$	mixed object: hyperbolic and reciprocal parts
Bose–Einstein	$1/(e^x-1)$	not a bounded binary share; outside the framework

Two remarks on the table. The ratio $I(1)/I(-1) = 1/8$ of the altitude integrals, highlighted in Revision 2 as holding "across all three descriptions," holds simply because a definite integral does not depend on the variable used to compute it. And "mixed object" should be read carefully: on a closed budget everything is a function of the single state χ, so a mixed object is one that is *compactly written* only as a combination of quantities from different coordinates, as with the black-hole recoil of Section 9.

8 Instantiation case studies

Special relativity (rapidity, closed, oriented). With signature $(-,+)$, for a timelike displacement, the null components $u = c\,dt + dx$, $v = c\,dt - dx$ give the partition $a = u/(u+v) = (1+\beta)/2$. Then $h^2 = ab = -ds^2/(4c^2dt^2)$, so $2h = d\tau/dt$, and the odds a/b equal the squared Bondi factor $k^2 = e^{2\xi}$. Einstein addition multiplies odds. Embedding, native coordinate and orientation are all supplied by the mechanism; the Fisher angle is available but is not boost-invariant [10].

Bernoulli estimation (Fisher, closed, unoriented). Here the Fisher metric is the physically relevant one, the Cramér–Rao bound reads $\text{Var}(\hat{p}) \geq h^2/n$ for n trials, and the angle $\arcsin\sqrt{\hat{p}}$ has approximately constant variance $1/(4n)$.

Flat ΛCDM (rapidity in e-folds, closed). With $a = \Omega_\Lambda$, $b = \Omega_m$ and $N = \ln(\text{scale factor})$, the ratio Ω_Λ/Ω_m grows as e^{3N}, so $\chi = \tanh\left(\frac{3}{2}N + c\right)$ and

$$\frac{d\xi}{dN} = \frac{3}{2}, \qquad V^{(1)}(\chi) = -\tfrac{1}{2}\Big(\frac{d\chi}{dN}\Big)^2 \frac{1}{1-\chi^2} = -\tfrac{9}{8}(1-\chi^2) = -\tfrac{9}{2}h^2. \tag{11}$$

(With the opposite orientation, $\chi \to -\chi$ and $d\xi/dN = -\frac{3}{2}$.) In e-fold time the expansion is linear in rapidity, consistent with the constant-rapidity result of [4]. More generally, for any two non-interacting fluids with constant equations of state, $d\xi/dN = \frac{3}{2}(w_b - w_a)$. The same extraction with the scale factor itself as coordinate gives the skewed shape $\propto (1-\chi)^{5/3}(1+\chi)^{1/3}$ (the form $(1-\chi)^{1/3}(1+\chi)^{5/3}$ of [1] corresponds to the opposite orientation $a = \Omega_m$). The difference is a choice of coordinate, and composition (e-folds add) selects N. This is a normal form of known cosmology, not new evidence.

Damped oscillator (closed internally, leaky globally). For T and V of an oscillator, $\ell = (T - V)/(T + V)$ and $2h$ satisfy $\ell^2 + (2h)^2 = 1$; for the harmonic oscillator the angle advances uniformly with phase [3]. With damping, the dissipated energy is the leakage channel and the state spirals inward in the half-disk.

Informative failures. Gauge-coupling running admits no autonomous one-dimensional potential, because the extracted law depends on the absolute scale (Section 5). For the reflected pair of real parts $(\frac{1}{2} + \delta, \frac{1}{2} - \delta)$ of a hypothetical off-line Riemann zero, the displacement acts on the explicit-formula side as an exponential reweighting $e^{-\delta A}$ whose composition is additive in δ itself, not in the rapidity artanh(2δ); translating it into rapidity is therefore a true mismatch [21]. In both cases the procedure correctly says "do not read the arc this way."

9 Case study: the SXS binary-black-hole catalog

Data and plan. We used the SXS catalog metadata [22, 23], release v3.0.0 (4148 binary-black-hole simulations). We kept non-deprecated, quasi-circular runs (reference eccentricity below 0.01) with $0.001 < E_{\rm rad}/M < 0.2$, leaving $N = 3527$, of which 81 are non-spinning. The instantiation and the tests were written down before fitting (`sxs_PREREGISTRATION.md`).

Instantiation (fixed from the physics).

- *Embedding*: $a = m_1/M$, $b = m_2/M$, the mass shares at the reference time ($m_1 \geq m_2$, so orientation is a labelling convention here).
- *Closure*: not closed through merger. The leakage fraction is $\lambda = E_{\rm rad}/M = 1 - M_f/M$, and the remnant fraction $1 - \lambda$ is the radius of Section 5.
- *Coupling*: $h^2 = ab = \eta$, the symmetric mass ratio.
- *Native coordinate (prediction)*: η, the expansion variable of post-Newtonian theory.

Results (Tier 2).

1. **$\eta = h^2$ gives the most compact description (compression test).** For the 81 non-spinning runs (21 of them equal-mass), polynomials of degree 1–3 in each candidate coordinate were scored by 5-fold cross-validation, repeated 200 times. At degree 2, the root-mean-square error for $E_{\rm rad}/M$ ($\times 10^{-4}$, mean with standard deviation over repeats) is η 1.96 (0.10); h 7.1 (0.7); χ 9.7 (0.3); Fisher angle 16.1 (0.7); rapidity 19.5 (0.9); R 73 (26). For the final spin, η gives 4.10 (0.23) against 13.5 (1.3) for the next best. At degree 3 the leading coordinates converge (η 0.99, χ 1.00 for $E_{\rm rad}/M$; h 2.24, η 2.30 for the spin), as monotone reparameterizations with enough terms must; at degree 1 the Fisher angle edges out η for $E_{\rm rad}/M$ (18.2 against 20.9). So the claim is one of compression: at the first order where any coordinate describes the data well, the coordinate chosen from the mechanism does so with several times smaller error. The Fisher and rapidity readings are not native here.

2. **The leakage law has an independent anchor, but these data do not pin it.** In the test-particle limit the radiated energy is $E_{\rm rad}/M = (1 - \sqrt{8/9})\,\eta + O(\eta^2)$, where $1 - \sqrt{8/9} = 0.0572$ is the binding energy of the innermost stable circular orbit (ISCO) of a Schwarzschild black hole [24, 25]: *leakage* $\approx$ *(ISCO binding energy)* $\times\, h^2$ at small coupling. (Here $1 - M_f/M$ includes the small initial binding energy, so it estimates the energy radiated from infinite separation, which is the quantity the test-particle limit describes.) The smallest η in the sample is 0.045, and the extrapolated $O(\eta)$ coefficient depends on the fit: 0.060 from a local linear fit of $E_{\rm rad}/(M\eta)$ for $\eta < 0.12$, 0.046 from a global quadratic, 0.069 from a global cubic. The data are consistent with the test-particle value but do not test it sharply.

3. **The recoil is compactly written as coupling times asymmetry.** The kick must vanish for equal masses (by symmetry) and in the test-particle limit, and it is odd in χ as a vector. The Fitchett–González form [26, 27], which builds in both zeros,

$$v = A\,\eta^2\sqrt{1-4\eta}\,(1 + B\eta) = A\,h^4\,\chi\,(1 + Bh^2)$$

with two parameters outperforms every unconstrained single-coordinate cubic (four parameters) in cross-validation (2.07 against 2.43, in units of $10^{-5}c$). Part of that advantage is the built-in endpoint zeros, which is domain knowledge of exactly the kind an instantiation is meant to supply. The fit gives $A = 1.31 \times 10^4$ km/s and $B = -1.34$, with a maximum of 173 km/s at $\eta = 0.192$; the published values are $A = 1.20 \times 10^4$ km/s, $B = -0.93$, and 175 ± 11 km/s at $\eta = 0.195$ [27].

4. **The Revision-2 equipartition differential does not replicate** (Section 10).

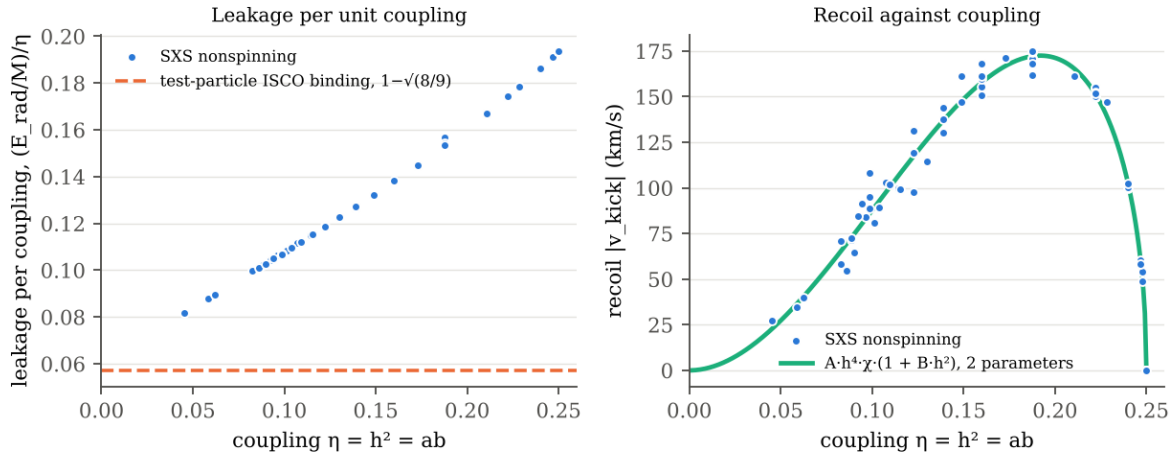


Figure 1: Non-spinning SXS runs. Left: leakage per unit coupling, $(E_{\rm rad}/M)/\eta$, against $\eta = h^2$, with the test-particle ISCO binding energy $1 - \sqrt{8/9}$ (the $\eta \to 0$ limit) for comparison; the smallest η in the sample is 0.045. Right: recoil against η with the two-parameter coupling-times-asymmetry form.

What this shows. Run as a procedure — fix the instantiation from the mechanism, then test — the framework recovers known physics in partition language and identifies which reading of the arc the domain uses. It does not supply new black-hole physics, and nothing here claims it does.

10 Withdrawn and corrected material

The equipartition differential (withdrawn). Revision 2 reported that the variance explained by partition coordinates grows with distance from equipartition (SDSS density: 48-fold, Pearson +0.86; SXS: 13.7-fold, +0.82; SDSS gradient margin: +0.87). Two checks show this should not be relied on.

1. *Null simulation.* With the SDSS bin edges of Revision 2 (0, 0.05, 0.10, 0.15, 0.20, 0.30, 0.50, 0.99), we simulated observables with no special role for asymmetry (script `equipartition_null_test.py` in the reproducibility files). An observable that depends on χ with the same strength everywhere gives an 88-fold increase in partial R^2 (Pearson +0.94); one depending on h gives 9.8-fold (+0.84); one depending on $\sin 2\chi$ gives 2.0-fold (+0.27). The main cause is the unequal bin widths: the last bin is about ten times wider than the first, and within-bin predictor variance grows with the square of the width. Basis-preference margins also grow with χ under these nulls (Pearson up to +0.91), because all candidate bases agree to leading order near $\chi = 0$. With only seven bins, a Pearson coefficient has a 95% interval of roughly ± 0.7. For the SXS bins of Revision 2 (0, 0.1, ..., 0.65, 0.95), the width ratio is only 3, so bin width alone accounts for at most part of the published SXS pattern; that claim fails on replication instead.
2. *Replication.* On the current SXS catalog with the published bins, the ratio is 1.8 and the Pearson correlation −0.17; with equal-width bins, 1.6 and +0.11.

The SDSS results could not be re-tested here. In view of the null simulation, they are withdrawn pending a re-analysis with equal-width bins and matched null controls. The corresponding checklist item of Revision 2 is reversed (Section 12).

Removed heuristics. The "dimensional selection" of Revision 2 (a one-parameter family of metrics tuned so that one member reproduces an inverse-square law) fits a family to a known target, which is the threshold-manufacturing pattern of [4]. The "six Pythagorean powers" classification had no testable content. The "infinite half-step" remark restated the boundary behaviour of Remark 1. All three are removed. The claimed pendulum and Kepler dynamics under a shared potential are also removed: dynamics are supplied by the domain, not by the choice of kinetic metric [1].

Corrected translation. For the generating function, Revision 2 gave $2\,\mathrm{gd}(\xi)\cosh\xi$ (rapidity) and $2\sqrt{2y}\arctan\sqrt{2y-1}$ (radial). These are the forms of θ/h, not of $G = \varphi/h = (\mathrm{gd}\,\xi + \frac{\pi}{2})\cosh\xi$: they used the Fisher angle θ where $2\varphi = \theta + \frac{\pi}{2}$ is required. They vanish at equipartition, where $G = \pi/2$. The corrected forms are in Appendix A.

11 Safeguards

1. **The arc is automatic.** Every two-channel budget lies on it (Proposition 1); recurrence across domains is expected, not evidence.
2. **Chart contributions.** Divergences of R, ξ or y at $|\chi| \to 1$ come from the coordinates. In extracted potentials, the factor $(1-\chi^2)^{-1}$ is the chart and the domain physics lives in the numerator [1].
3. **Monotone equivalence.** All the coordinates are monotone functions of one another, so rank-based statistics cannot tell them apart, and differences under polynomial fits reflect curvature, not new information.
4. **Native composition.** A coordinate is native only if the domain's composition law is additive in it, or its dynamics close in it.

5. **Orientation.** If the mechanism distinguishes the channels, use signed χ or ξ; h, R and y are blind to orientation.
6. **Closure.** Ask whether $a + b$ is conserved. If it is not, track the radius, and remember that R_{act} grows with leakage alone.
7. **Declare the metric.** Any statement about distances or regions must say which metric is meant (Remark 1).
8. **Pre-commitment.** Fix the instantiation before comparing with data or targets.
9. **Normal forms are not tests.** Report an over-determination test before claiming content.

12 Practical checklist

1. Identify the two channels and write the instantiation (Definition 1) before looking at data.
2. Check closure; if the budget leaks, measure L independently.
3. Choose the native coordinate by composition or dynamics, not by appearance.
4. Translate with Table 1; check that domains, bounds and singularities correspond.
5. Classify: exact translation, partial correspondence, mixed object, or mismatch.
6. For empirical claims, use equal-width bins or binless methods, compare against a shuffled or constant-effect null, and use held-out data.
7. Run at least one over-determination test and report it, including failures.

13 Epistemic tiers

- **Tier 1**: Propositions 1–3; the metrics, the inversion identity and the Gudermannian dictionary; the exponential-family duality; the worked translations of Table 3; the e-fold form of the ΛCDM partition. All are checked, symbolically or numerically, in `verify_rev3.py`.
- **Tier 2**: the SXS results of Section 9; the withdrawal analysis of Section 10.
- **Tier 3**: the constraint-type heuristic of Table 2; any claim that two domains share physics because they share a reading of the arc.

14 Conclusion

The binary partition gives every two-channel budget the same arc, and the dictionary of Section 3 translates exactly between the coordinates people use on it. That much was in Revision 2. What Revision 3 adds is a discipline for using it. The domain instantiates the arc — what the channels are, whether the budget is closed, whether the coupling is realized, which coordinate composition makes additive, whether orientation matters — and it must do so before the comparison. Translations of known formulas are normal forms. Claims need over-determination tests, and failed tests classify domains rather than refuting the arc.

The SXS analysis shows what this looks like in practice: the coordinate predicted from the physics gives the most compact description out of sample, the leakage law has an independently known limit that the data are consistent with, and the recoil is compactly written as coupling times asymmetry once the symmetry-required zeros are built in. The empirical pattern that Revision 2 relied on is withdrawn, because a simple null reproduces it. The guide is smaller as a result, and more useful.

Reproducibility

`verify_rev3.py` checks the Tier-1 identities symbolically or numerically. `equipartition_null_test.py` reproduces the null simulation. `load.py`, `analyze.py` and `cv_repeat.py`, with the plan in `sxs_PREREGISTRATION.md`, reproduce Section 9 from `simulations.json` of release v3.0.0 of the SXS catalog data (`sxs-collaboration/sxscatalogdata`), which `fetch_data.py` downloads and checks against a fixed checksum. All files are in the folder `two-face-problem-rev3/` of the repository `https://github.com/innerlightr-wq/constraint-geometry-domain-physics`.

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A Worked translations in full

Fermi–Dirac and logistic. Take a to be the vacant fraction and $b = f$ the occupied one. Then $f(x) = 1/(e^x + 1) = (1 - \tanh\xi)/2$ with $\xi = x/2$, $a = (1 + \tanh\xi)/2$, and $\log\frac{a}{b} = 2\xi = x$; with the opposite assignment, $\xi \to -\xi$. In the Fisher angle, $f = (1 - \sin\theta)/2$ with $\theta = \mathrm{gd}(x/2)$; in the radial coordinate, $f = (1 - \sqrt{1 - 1/(2y)})/2$ with $y = \frac{1}{2}\cosh^2(x/2)$.

Mercator. Latitude ϕ_{lat} maps to the Mercator ordinate $\psi = \ln(\sec\phi_{\mathrm{lat}} + \tan\phi_{\mathrm{lat}}) = \mathrm{gd}^{-1}(\phi_{\mathrm{lat}})$, and $\psi = \operatorname{arccosh}(\sec\phi_{\mathrm{lat}})$ for $\phi_{\mathrm{lat}} \geq 0$.

Bernoulli Fisher information. $I_F(p) = 1/(p(1-p)) = 1/h^2 = 4R = 8y = 4\sec^2\theta = 4\cosh^2\xi$.

Aberration. With $\beta = \tanh\xi$ and the convention $\cos\theta' = (\cos\theta + \beta)/(1 + \beta\cos\theta)$, $\tan(\theta'/2) = \sqrt{(1-\beta)/(1+\beta)}\,\tan(\theta/2) = e^{-\xi}\tan(\theta/2)$, and $e^{-\xi} = \sqrt{2y} - \sqrt{2y-1}$.

VSWR. $\mathrm{VSWR} = (1 + |\Gamma|)/(1 - |\Gamma|) = e^{2|\xi|} = (\sqrt{2y} + \sqrt{2y-1})^2$. For a resistive load, $\Gamma = (Z - Z_0)/(Z + Z_0) = \tanh(\frac{1}{2}\ln(Z/Z_0))$, so $\xi = \frac{1}{2}\ln(Z/Z_0)$ and $\mathrm{VSWR} = \max(Z/Z_0, Z_0/Z)$.

Altitude integrals. $I(\alpha) = \int_0^1 [x(1-x)]^{\alpha/2} dx = B(\frac{\alpha}{2}+1, \frac{\alpha}{2}+1)$; $I(1) = \pi/8$, $I(-1) = \pi$. In rapidity, $I(\alpha) = 2^{-(\alpha+1)}\int_{-\infty}^{\infty} \mathrm{sech}^{\alpha+2}\xi\, d\xi$.

Generating function (corrected). Since $\int_0^1 h^{2m} dx = B(m+1, m+1)$,

$$G(t) = \sum_{m\geq 0} t^m \int_0^1 h(x)^{2m} dx = \int_0^1 \frac{dx}{1 - t\,x(1-x)} = \frac{4\arcsin(\sqrt{t}/2)}{\sqrt{t(4-t)}}.$$

Setting $a = t/4$, $\varphi = \arcsin\sqrt{a}$ and $h_a = \sqrt{a(1-a)}$ gives $G = \varphi/h_a$. Since $\varphi = \frac{1}{2}(\theta + \frac{\pi}{2})$ and $h = \frac{1}{2}\,\mathrm{sech}\,\xi$,

$$G = \big(\mathrm{gd}\,\xi + \tfrac{\pi}{2}\big)\cosh\xi = \sqrt{2y}\,\big(\pm\arctan\sqrt{2y-1} + \tfrac{\pi}{2}\big),$$

with the sign of $\chi = 2a - 1$. At equipartition $G = \pi/2$. Revision 2 gave $2\,\mathrm{gd}(\xi)\cosh\xi$ and $2\sqrt{2y}\arctan\sqrt{2y-1}$, which are the forms of θ/h, not of G.

Langevin and Bose–Einstein. $\coth x - 1/x$ combines a hyperbolic term and a reciprocal term, and no single coordinate makes it simple. $1/(e^x - 1)$ is unbounded and is not a share of a two-channel budget.

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